

A PROJECT REPORT ON

PolicyVault Nexus

Responsible Data Sharing in Fintech Environment

Submitted towards the
Partial Fulfilment of the Requirements of

BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
By

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CERTIFICATE

This is to certify that the project report entitled

PolicyVault Nexus
Responsible Data Sharing in Fintech Environment

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is a bonafide work carried out by them under the supervision of Dr. S. A. Khaire and it is approved for the partial fulfilment of the requirement of Bachelor of Technology (Artificial Intelligence and Data Science) during academic Year 2025-2026.

This project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

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Abstract

The exponential growth of fintech ecosystems has ushered in unprecedented innovation in digital finance—enabling real-time customer services, cross-platform integrations, and data-driven personalization. However, this rapid evolution has come at the cost of increased privacy risks, regulatory exposure, and eroding customer trust, especially when sharing sensitive personal and financial data across distributed entities and APIs. This project proposes PolicyVault Nexus: a comprehensive, privacy-by-design, ephemeral data gateway architecture that enables real-time, RBI and DPDP-compliant financial data sharing while strictly minimizing exposure and abuse risks. Key components include in-memory-only data processing (zero storage), Format Preserving Encryption (FF3-1) for field-level tokenization, automated policy enforcement (Policy-as-Code and OPA), dynamic, user-centric consent artifacts compatible with regulatory frameworks (GDPR/DPDP/RBI-AA), and continuous, auditable access monitoring. PolicyVault Nexus eliminates persistent sensitive storage, supports granular user controls, and ensures privacy preservation even after data egress to third-party partners. By combining cryptographic innovation, real-time ephemeral computing, and automated compliance with regulatory consent models, this framework achieves scalable, transparent, and user-first privacy protection—restoring trust and enabling responsible fintech innovation in highly distributed ecosystems.

Keywords: *Privacy-by-Design, Ephemeral Computing, Format Preserving Encryption, FF3-1, Tokenization, Zero Storage, RBI Account Aggregator, DPDP Act, GDPR, Consent Artifacts, Policy-as-Code, Open Policy Agent, Dynamic Consent Management, Real-Time Data Sharing, Fintech Data Privacy, Audit and Compliance, User Control, Data Minimization, Blockchain, Financial Data Security*

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CHAPTER 1

INTRODUCTION